



GLOBALTECHTM

Expanding Globally by Intelligent Solutions

ProSYS

6 – Axis + Galvo Motion Control Card

USER MANUAL

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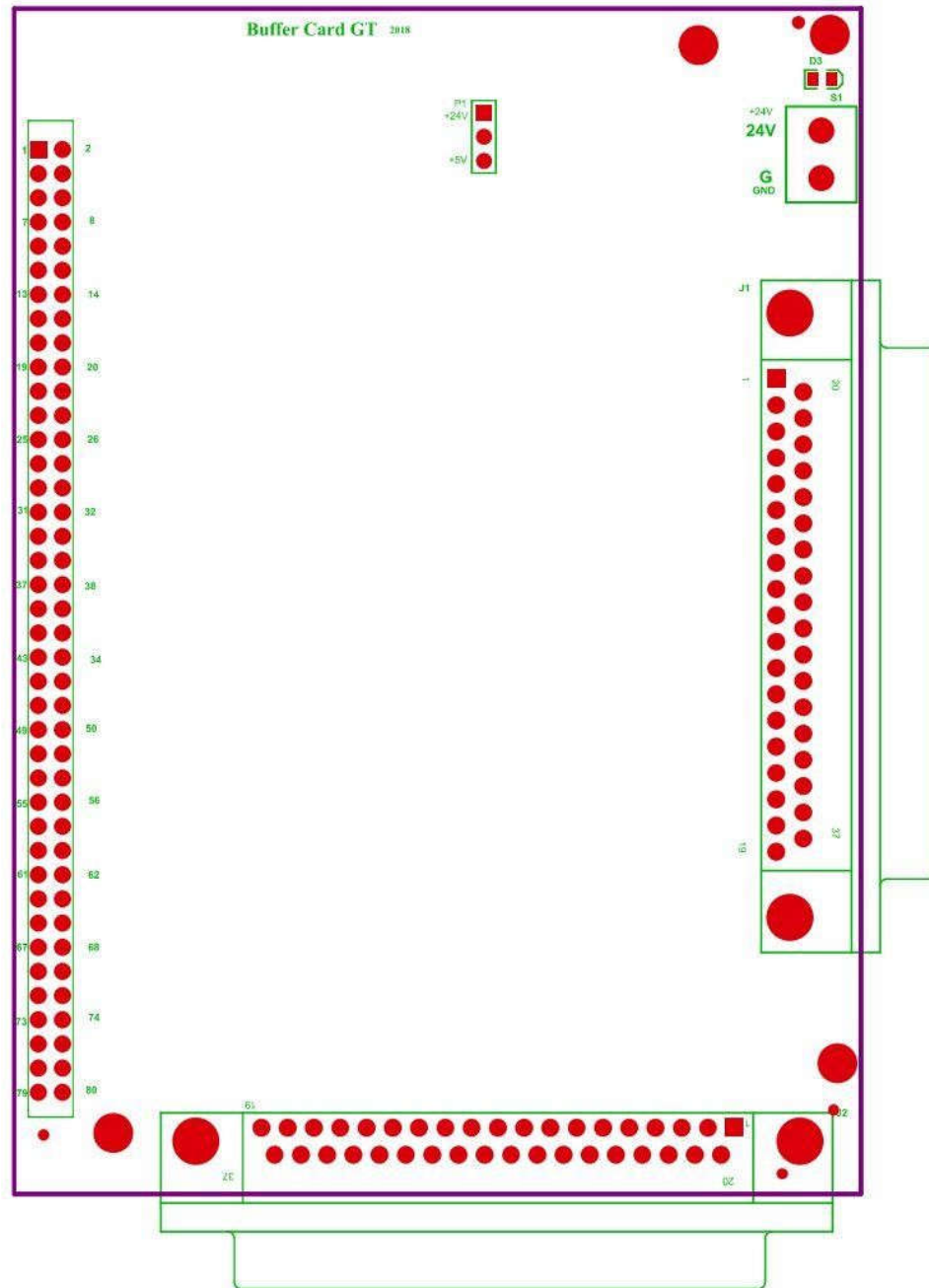
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Introduction

GT PROSYS V10 is 5 Axis Stepper or Servo motor motion control card with USB 3.0.

Features

- USB3.0 Communication
- Generate Maximum Pulse output rate of 6.5 MPPS
- Motion profile: Trapezoidal
- Acceleration/Deceleration time adjustable
- On-the-fly speed change or position change
- Any 3 of 5 axes linear interpolation motion
- Any 2 of 5 axes circular or arc interpolation motion
- Any 2 of 5 axes circular or arc interpolation motion
- Continuous buffered move
 - 1,2 or 3 axes linear interpolation
 - Any two axis circular or arc interpolation
 - Digital output ON/OFF
- Pulse/Dir line driver output
- Software limit function for position and speed
- Origin signal input, positive and negative end limit switch input and alarm signal input for 5 axes.
- Opto-isolated I/O: General purpose eight Digital inputs and Open collector outputs
- Software support multiple card
- Position input range (Pulse) 28 bit (-134,217,728 ~ +134,217,727)
- Position counter 32 bit signed (-2,147,483,648 ~ +2,147,483,647)
- Galvo motion with speed control galvo position (0 to 65535)



Isolation Board V10

DB37 Female connector Pin Assignment

Connector J1

Pin	I/O	Name	Pin	I/O	Name
1	O	X-Pulse(+)	20	I	Z-ALM
2	O	X-Pulse(-)	21		GND1
3	O	X-Dir(+)	22	O	U-Pulse(+)
4	O	X-Dir(-)	23	O	U-Pulse(-)
5	I	X-ORG	24	O	U-Dir(+)
6	I	X-ALM	25	O	U-Dir(-)
7		GND1	26	I	U-ORG
8	O	Y-Pulse(+)	27	I	U-ALM
9	O	Y-Pulse(-)	28		GND1
10	O	Y-Dir(+)	29	O	V-Pulse(+)
11	O	Y-Dir(-)	30	O	V-Pulse(-)
12	I	Y-ORG	31	O	V-Dir(+)
13	I	Y-ALM	32	O	V-Dir(-)
14		GND1	33	I	V-ORG
15	O	Z-Pulse(+)	34	I	V-ALM
16	O	Z-Pulse(-)	35		GND1
17	O	Z-Dir(+)	36	O	24V
18	O	Z-Dir(-)	37	O	5V
19	I	Z-ORG			

Connector J2

Pin	I/O	Function	Pin	I/O	Function
1	I	W-Pulse(+)	20	O	DO2
2	I	W-Pulse(-)	21	O	DO3
3	I	W-Dir(+)	22	O	DO4
4	I	W-Dir(-)	23	O	DO5
5	I	W-ALM	24	O	DO6
6	I	W-ORG	25	O	DO7
7	I	U-EL(+)	26	O	DO8
8	I	U-EL(-)	27	O	5V
9	I	V-EL(+)	28		GND1
10	I	V-EL(-)	29	I	VCC2
11	I	DI1	30		GND2
12	I	DI2	31	I	VCC3
13	I	DI3	32		GND3
14	I	DI4	33	O	PWM
15	I	DI5	34		GND1
16	I	DI6	35	O	ANALOG
17	I	DI7	36		GND1
18	I	DI8	37		NC
19	O	DO1			

Pulse	Clock output
Dir	Direction output
ORG	Origin signal or Home signal input
ALM	Alaram signal input
EL(+)	End limit Plus signal input
EL(-)	End limit Minus signal input
DI	General purpose digital input
DO	General purpose digital output
PWM	PWM Output
ANALOG	Analog Output Voltage
GND1	External power supply ground
VCC2	External power supply require for Digital Output (5V to 24VDC)
GND2	Ground for VCC2
VCC3	External power supply require for End limit and digital input (5V to 24VDC)
GND3	Ground for VCC3

**Galvo Isolation
Board V10
DB15 Female connector Pin Assignment**

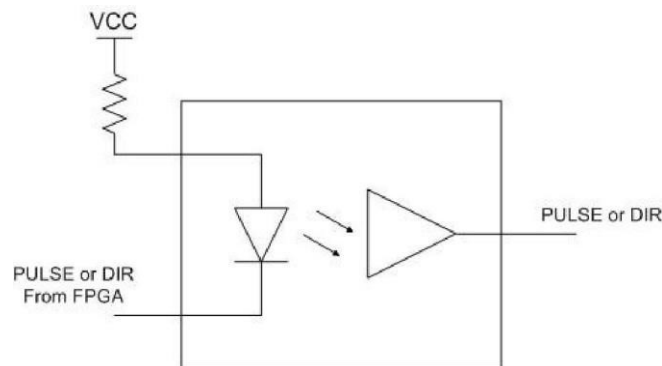
Connector J1(Galvo)

Pin No.	I/O	Name	Pin No.	I/O	Name
1	O	CLK (-)	9	O	CLK (+)
2	O	SYNC(-)	10	O	SYNC(+)
3	O	X (-)	11	O	X (+)
4	O	Y (-)	12	O	Y (+)
5		NC	13		NC
6		NC	14		NC
7		NC	15		NC
8		NC			

Signal Connection

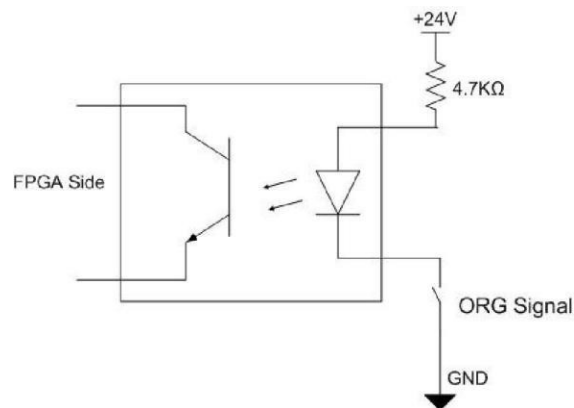
Pulse and Direction

Every axis has pulse and direction pin. Pulse pin gives output of pulse train and direction pin indicates direction. In servo motor drive, connect pulse output of card to PULSE+ and direction output to DIR+. Connect PULSE- and DIR- of drive to GND. Change drive parameter to signal PULSE/DIR, if required.



Origin or Home Signal (ORG)

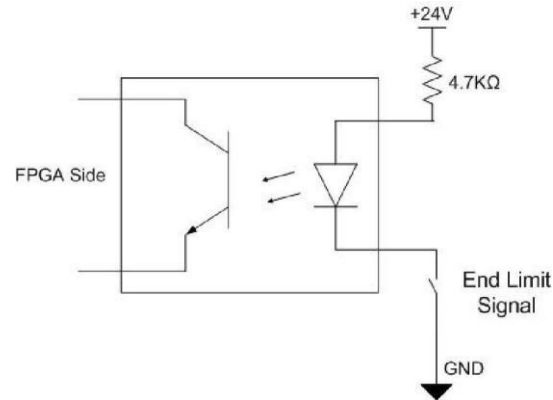
Home or ORG signal normally used to indicate home or origin position of mechanism. There is ORG input available for each axis. Connection diagram for origin or home (ORG) input is given below.



When card is operated with home function, then only card will detect home or origin signal. At detection of origin signal, motor will stop after slow down or immediately stop as per selection of mode. User can change origin signal detection logic (active high/low) by calling function.

End limit signal (EL+/EL-)

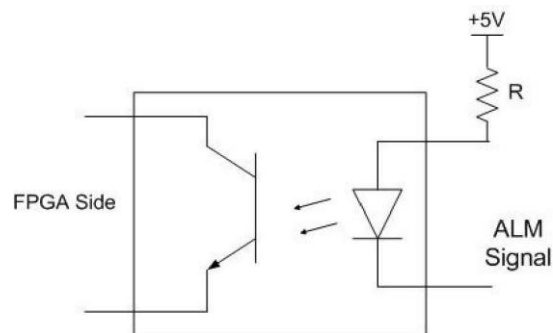
There are two end limit signal for each axis. One is ELP for positive end limit and ELM for negative end limit. Connection diagram for end limit (EL) input is given bellow.



User can enable or disable end limit by calling function. At detection of end limit signal motor will stop after slow down or immediately stop as per selected mode. User can change end limit signal detection logic (active high/low) by calling function.

Alarm Signal (ALM)

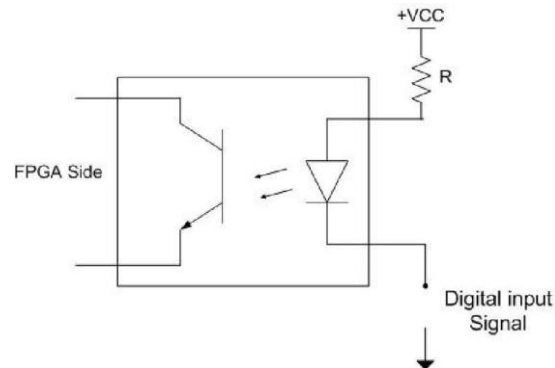
The alarm signal used to indicate status of servo drive. Connection diagram for alarm (ALM) input is given bellow.



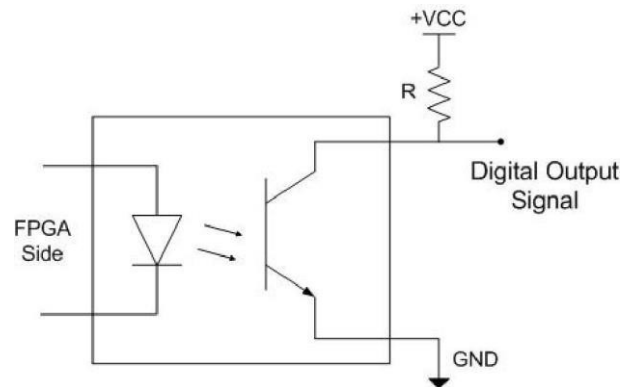
The ALM signal available in servomotor drive, which is normally Open collector type. To drive ALM signal minimum 5mA current require. ALM detection logic (active high/low) can change by calling function. At detection of ALM signal motor will stop after slow down or immediately stop as per selected mode.

General Purpose Digital Input / Output signal

There are 8 general purpose digital input / output signal. Input connection given bellow.



Digital output connection is given below.



Continuous Motion with Maximum Velocity

The command described here is for continuous motion of one axis. Output pulse starts from starting velocity and accelerates to specified maximum velocity. The function **GT_PROSYS_U3_tv_move()** is used to accelerate as trapezoidal motion profile. The output pulse is generated until new command is set or stop command is issued. User can change speed during motion using **GT_PROSYS_U3_v_change()**. The command **GT_PROSYS_U3_fix_speed_range()** is used to fix speed of motion.

For Example:-

GT_PROSYS_U3_tv_move(0, 10000, 50000, 0.1, 0)

By calling this function, 0 no axis(X-Axis) motor start motion with starting velocity 10000 pulse per second(PPS), accelerate to maximum velocity 50000 PPS in 0.1 second with trapezoidal motion profile. Motion of motor does not stop until **GT_PROSYS_U3_sd_stop()** or **GT_PROSYS_U3_abort()** function is called or issued.

GT_PROSYS_U3_v_change(0, 100000, 0.1)

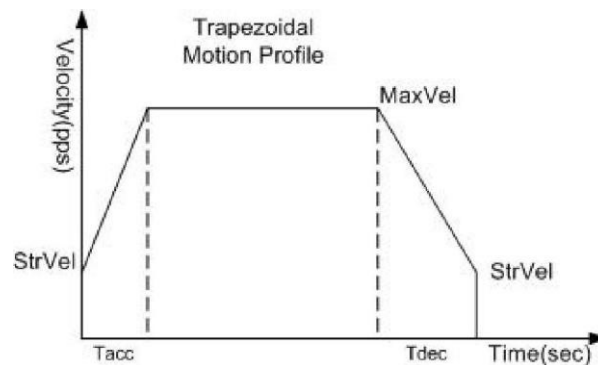
By calling this function, 0 no axis(X-Axis) motor current speed change to speed of 100000 PPS in 0.1 second. Those change or stop functions follow the same velocity profile as its original move functions, tv_move or sv_move.

Related Functions:-

**GT_PROSYS_U3_tv_move(),GT_PROSYS_U3_v_change(),
GT_PROSYS_U3_sd_stop(),GT_PROSYS_U3_abort(),
GT_PROSYS_U3_fix_speed_range(),GT_PROSYS_U3_enable_speed_limit(),
GT_PROSYS_U3_disable_speed_limit()**

Trapezoidal Motion Profile

The functions described here use to move from current position to specified position. The absolute and relative motion can be performed. In absolute mode, target position is assigned and in relative mode target displacement is assigned. Following diagram shows trapezoidal motion profile.



GT_PROSYS_U3_axis_move function is used for trapezoidal motion profile and relative mode or absolute mode.

The displacement or position must be given in unit of pulse. The starting velocity and maximum velocity must be positive and in pulse per second. Acceleration and deceleration time should be in unit of second. User can give different time for acceleration and deceleration. The function **GT_PROSYS_U3_motion_done()** is used to check the current status of axis, is axis in motion or not?.

For example:-

GT_PROSYS_U3_axis_move (0, 1, AxisArray*, DistArray*,10000,20000, 0.1)

X-axis current position = 10000 pulse

AxisArray[0] = 0

DistArray[0]=10000

By calling this function, 0 no axis(X-axis) motor start motion with starting velocity of 10000 PPS and form current position 10000, accelerate to maximum velocity of 20000 PPS in 0.1

second and move at constant speed of 20000 PPS. Start deceleration from maximum velocity 20000 PPS to starting velocity in 0.1 second and reach position and stop motion. Motion profile is trapezoidal. X-axis new position is 20000 (10000 + 10000). To do motion in reverse direction, enter distance or position value in negative in DistArray.(-10000)

GT_PROSYS_U3_axis_move (1, 1, AxisArray*, DistArray*,10000,20000, 0.1)

Z-axis current position = 100000 pulse

AxisArray[0] = 2

DistArray[0]=50000

By calling this function, 2 no axis(Z-axis) motor start motion with starting velocity of 10000 PPS and from current position 100000, accelerate to maximum velocity of 20000 PPS in 0.1 second and move at constant speed of 20000 PPS. Start deceleration from maximum velocity 20000 PPS to starting velocity in 0.1 second and reach position and stop motion. Motor start motion in reverse direction because current position is 100000 pulse, new absolute position is 50000. So motor moves 50000 pulse in reverse direction to reach position of 50000 pulse.

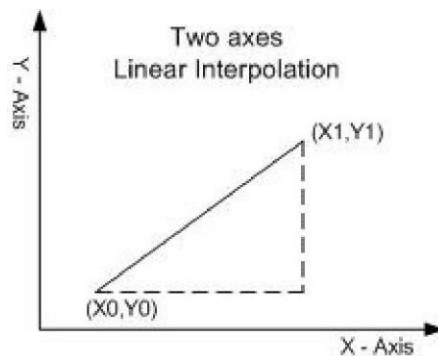
Related Function:-

GT_PROSYS_U3_motion_done()

Linear Interpolation of any 2 or 3 axes out of 5

Linear interpolation of any two or three axes means two or three axes start motion at same time and reach specified position at same time. Linear means the ratio of speed of every axis is a constant value.

2 Axes Linear Interpolation



For example:-

GT_PROSYS_U3_axis_move (0, 2, AxisArray*, DistArray*,1000,2500,0.1)

AxisArray[0] = 0

AxisArray[1] = 1

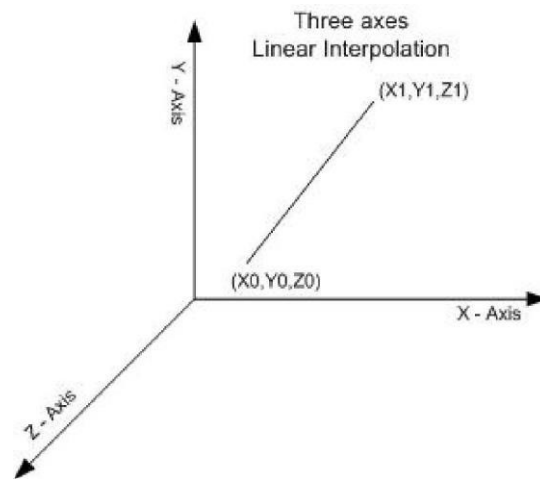
DistArray[0]=50000

DistArray[1]=10000

In this function of 2 axes (X and Y) linear interpolation, 50000 is X- axis relative position in pulse, 10000 is Y- axis relative position in pulse, 1000 is starting velocity, 2500 is maximum velocity, and 0.1 is acceleration time in second. With these parameters, X and Y axes start motion at same time and reach specified position at same time.

3 Axes Linear Interpolation

Any 3 of 5 axes performs linear interpolation. 3 axes linear interpolation means move selected 3 axes from current position to new specified position at same time.



For Example:-

GT_PROSYS_U3_axis_move (0, 3, AxisArray*, DistArray*,1000,2500, 0.1)

AxisArray[0] = 0

AxisArray[1] = 1

AxisArray[2] = 2

DistArray[0]=50000

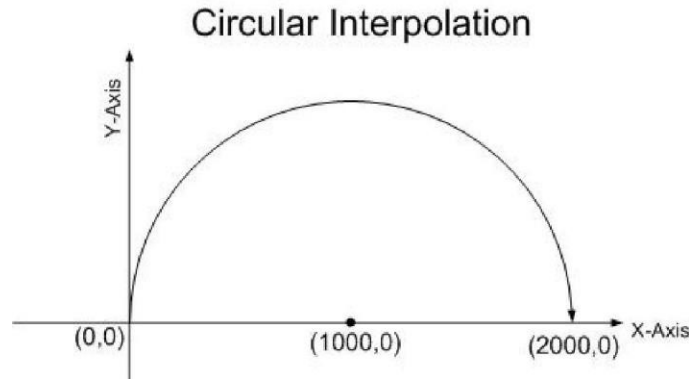
DistArray[1]=10000

DistArray[2]=75000

In this function of 3 axes (X, Y and Z) linear interpolation, to select 3 axes pass array, 50000 is X- axis relative position in pulse, 10000 is Y- axis relative position in pulse, 75000 is Z- axis relative position in pulse, 1000 PPS is starting velocity, 2500 PPS is maximum velocity, 0.1 is acceleration time in second. With these parameters, X, Y and Z axes start motion at same time and reach specified position at same time.

Circular Interpolation of Any 2 Axes

Any 2 of 5 axes can perform circular interpolation. In circular interpolation, selected two axes start simultaneously from initial position, and stop at end point. The path between them is circle or arc.



For example:-

GT_PROSYS_U3_axis_move (1, 4, AxisArray*, DistArray*,0,5000, 0.1)

AxisArray[0] = 0

AxisArray[1] = 1

DistArray[0]=1000 ,Cx

DistArray[1]=0 ,Cy

DistArray[2]=2000 ,Ex

DistArray[3]=0 ,Ey

DistArray[4]=0 ,Dir

In above function, first parameter 0 is for absolute position, second parameter 4 is for ARC, third parameter axis array is for which axis to move, fourth parameter Distarray is for Center ,Direction and End of Arc, fifth parameter 0 is starting velocity , sixth parameter 5000 is Maximum Velocity, last parameter 0.1 is Acceleration Time in Second.

Continuous motion or Buffer move

This is the function to perform continuous motion. The function for single axis, 2 axis interpolations, 3 axis interpolation, circular interpolation and digital output (DO) on/off can be used in this continuous motion.

To use continuous motion,

1. Call function **GT_PROSYS_U3_set_continous_mode_on()**. This function will set card in continuous motion mode. Before calling this function make sure, no axis is in motion. If any axis is in motion, wait for motion to complete for all axes.
2. Add motion commands in buffer. This command will store in computer buffer. You can get maximum buffer size by calling function **GT_PROSYS_U3_buffer_maxsize()**, You can also get current buffer size by calling function **GT_PROSYS_U3_get_buffer_size()**.
3. Check card buffer is ready to accept commands by calling function

GT_PROSYS_U3_buffer_ready(). If buffer is ready then send filled buffer for execution by calling **GT_PROSYS_U3_send_buffer()**.

4. After calling last function it is compulsory to call **GT_PROSYS_U3_set_continuous_mode_off()**. This function will cause the continuous motion to off.

GT_PROSYS_U3_buffer_pause(), this function will pause the motion.

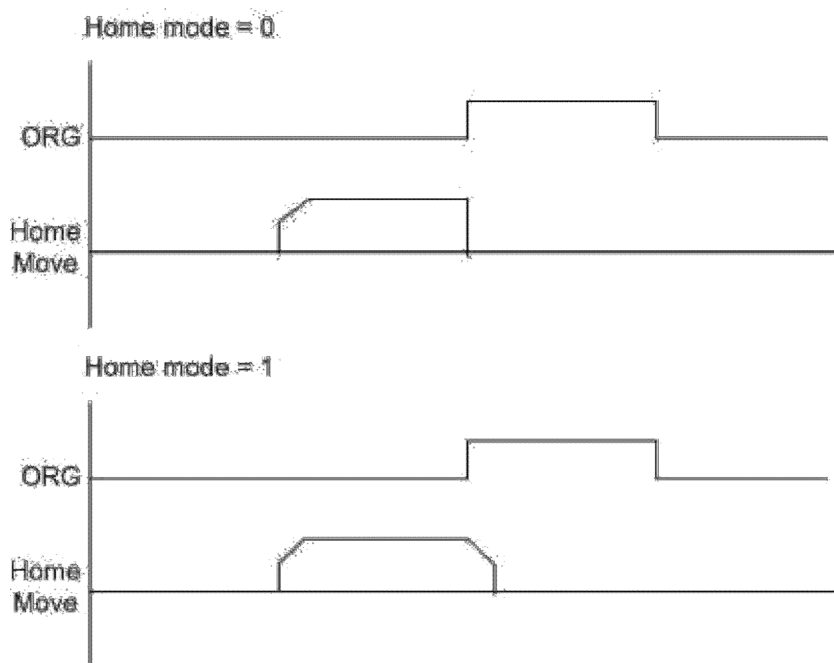
GT_PROSYS_U3_buffer_resume(), this function will resume the motion.

GT_PROSYS_U3_buf_abort(), this function will abort all motion, clear the buffer and exit from continuous motion.

GT_PROSYS_U3_get_buf_cur_id(), this function will return command number, which is currently in execution

Home Return Mode

Home return mode can be perform by using **GT_PROSYS_U3_home_axis()** function. **GT_PROSYS_U3_set_home()** function is used to set home mode and home signal logic. By calling this function, axis start motion in selected direction and search for home signal. When home signal is detected, axis stops immediately or stops after deceleration as per selected mode. **GT_PROSYS_U3_escape_home()** function is used to escape from home signal. In this function direction is selected. In escape home function, when origin signal is ON, the motion will last until origin signal is OFF. After performing home move or escape move function, user can set or reset axis position by using **GT_PROSYS_U3_set_command()** function.



For example:-

GT_PROSYS_U3_home_axis(0, 1000, 5000, 0.1, 0)

In this function, 0 is axis number(X axis), 1000 is starting velocity, 5000 is maximum velocity, 0.1 is acceleration time in second and 0 is positive direction.

GT_PROSYS_U3_set_command(0, 1000)

In this function, 0 is axis number (X-axis) and 1000 is position to be set.

Related function:-

GT_PROSYS_U3_home_axis(),

GT_PROSYS_U3_set_home(),

GT_PROSYS_U3_escape_home(),

GT_PROSYS_U3_set_command()

Alarm signal (ALM)

The ALM signal is an output from servomotor drive. Usually, it is designed to inform that something wrong with the drive or motor.

The ALM input pin receives alarm signal output from servo drive. When this signal is detects, axis motion stops immediately or stops after deceleration as per mode selected. If alarm signal is ON at starting, the particular axis will not move. User can change alarm signal detection logic and mode by **GT_PROSYS_U3_set_alm()** function.

For example:-

GT_PROSYS_U3_set_alm(0, 1, 0)

In this function, first parameter 0 is axis number, second parameter 1 is for logic, active high and third parameter 0 is for mode.

Alarm logic 0 – Active low (Default)

Alarm logic 1 – Active high

Mode 0 – Immediately Stop axis (Default)

Mode 1 – Stop axis after deceleration

Related function:-

GT_PROSYS_U3_set_alm()

End Limit (EL)

End limit is signal used to set hardware position limit. When EL is ON, axis will stop immediately or stop after deceleration as per selected mode. **GT_PROSYS_U3_set_el()** function is used to change ON\OFF logic and stop mode.

ELP signal indicates end limit for positive direction. ELM signal indicates end limit for negative direction. When axis moves towards positive direction and ELP signal is detected, axis

will stop moving and axis only moves towards negative direction. User can enable or disable EL signal using **GT_PROSYS_U3_enable_hard_limit()** and **GT_PROSYS_U3_disable_hard_limit()**. If EL signal is disable then ELP and ELM will not work as limit. It will not stop axis on detection of ELP or ELM signal.

For example:-

GT_PROSYS_U3_set_el(0, 1, 0)

In this function first parameter 0 is axis number, second parameter 1 is detection logic and third parameter 0 is mode.

Logic 0 – Active low (Default)

Logic 1 – Active high

Mode 0 – Immediately Stop axis (Default)

Mode 1 – Stop axis after deceleration

GT_PROSYS_U3_enable_hard_limit(0)

This function enable hardware limit for 0 number axis (X-axis).

Related function:-

GT_PROSYS_U3_set_el()

GT_PROSYS_U3_enable_hard_limit()

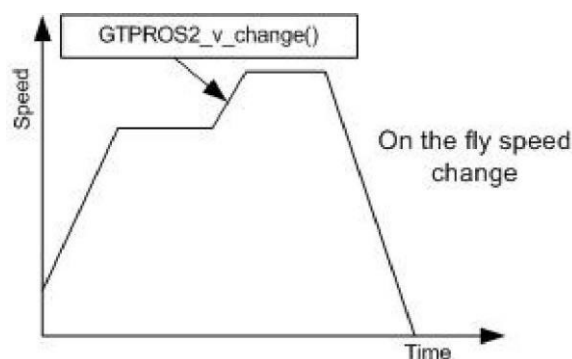
GT_PROSYS_U3_disable_hard_limit()

Change the speed or position on the fly

User can change axis speed while axis is in motion. User can also change axis position while axis is in motion.

On The Fly Speed Change

On the fly speed change applicable on single axis only.



For example:-

GT_PROSYS_U3_v_change(0, 50000, 0.1)

In this function, 0 is axis number(X-axis), 50000 is new velocity in pulse per second and 0.1 is acceleration or deceleration time in second. If current velocity of axis is greater than new velocity, axis will decelerate to new velocity in 0.1 second. If current velocity of axis is less than new velocity, axis will accelerate to new velocity in 0.1 second.

GT_PROSYS_U3_fix_speed_range(0, 150000)

In this function, 0 is axis number(X-axis), 150000 is speed limit in pulse per second. This function will only work when speed limit on particular axis is enabled.

Related function:-

GT_PROSYS_U3_v_change()

GT_PROSYS_U3_sd_stop()

GT_PROSYS_U3_abort()

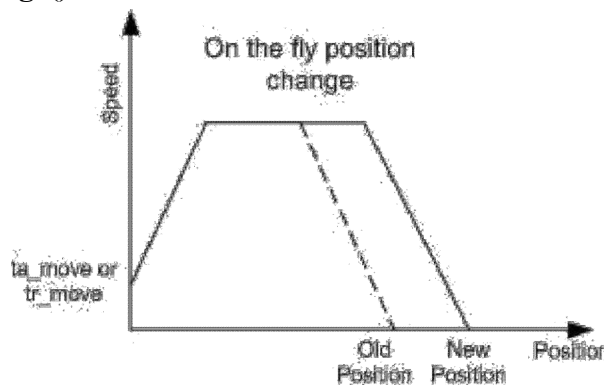
GT_PROSYS_U3_fix_speed_range()

GT_PROSYS_U3_enable_speed_limit()

GT_PROSYS_U3_disable_speed_limit()

On The Fly Position Change

User can change absolute position of axis when axis is in motion by calling **GT_PROSYS_U3_p_change()** function.



It is to change the target position defined originally by these two functions. After changing position, the axis will move to the new target position and totally forget the original position. If the new position is in passed path, it will cause the axis to decelerate to stop, than move in reverse direction. The acceleration and deceleration rate, the StrVel and MaxVel will keep the same as original setting. The position will be defined in this function, consider as absolute position.

For example:-

GT_PROSYS_U3_p_change(0, 500000)

In this function, 0 is axis number(X-axis), 500000 is new absolute position in pulse.

Related function:-

GT_PROSYS_U3_p_change()

Software Limit Function

There are 2 software limits for each axis, one for positive side and second for negative side. Software limits work same as hardware limit or physical limit switch. This function will pre-set position counter limits value. When position counter will reach limit value then axis will stop immediately or stop after deceleration. User can enable or disable software limit. Following functions are use for software limit.

GT_PROSYS_U3_set_soft_limit() – To set software limit
GT_PROSYS_U3_enable_soft_limit() – To enable software limit
GT_PROSYS_U3_disable_soft_limit() – To disable software limit

For example:-

GT_PROSYS_U3_set_soft_limit(0, 250000, -250000)

In this function, 0 is axis number (X-axis), 250000 is position counter limit for positive side and - 250000 is position counter limit for negative side.

Related function:-

GT_PROSYS_U3_set_soft_limit(),
GT_PROSYS_U3_enable_soft_limit(),
GT_PROSYS_U3_disable_soft_limit()

Test Application

The screenshot displays the 'GtMotion Test Utility' software interface. It features a top status bar with 'Acc. Time' (0.1) and 'Dec. Time' (0.1) fields, and speed settings for 'High Speed' (50000) and 'Low Speed' (10000). The main workspace is divided into five columns, each representing an axis: X-Axis, Y-Axis, Z-Axis, U-Axis, and V-Axis. Each axis column contains a set of controls including 'H/W Pos. Limit' (ON/OFF), 'S/W Pos. Limit' (+Ve/-Ve), 'Speed Limit' (On/Off), 'Move Count' (S-A, S-R), 'Read' (X, Y, Z, U, V), 'Reset' (X Count, Y Count, Z Count, U Count, V Count), 'Vel.' (0), 'Time' (0), 'OnFly Vel. Chg', 'OnFly Pos. Chg', 'STOP' (STOPX, STOPY, STOPZ, STOPU, STOPV), 'ABORT', 'HOME' (+, -, EscHM, [+], [-]), and 'JOG' (T, [+], T, [-], S, [+], S, [-]). To the right of the axis columns are additional settings: 'Position/Dist', 'Select Mode' (Relative/Absolute), 'Select Profile' (Trapazoid/S-Curve), 'Axis1', 'Axis2', 'Axis3' (X, Y, Z, U, V, None), 'Any 2 Axis', 'Any 3 Axis', 'XY', 'ZU', 'Select Direction' (Clockwise/Counter Clockwise), 'Select Mode' (Relative/Absolute), 'Center X' (1000), 'Center Y' (0), 'End X' (1800), 'End Y' (600), 'Velocity' (55000), 'Arc-Ang.2', 'XY', 'ZU', and a 'Buffer' button. At the bottom, there are 'Set DO', 'Read DO', 'Set', 'Reset', 'Bit Read', 'Bit Read DI', and 'BIT' controls, along with a status display for 'ALARM' and 'HOME' for each axis, and 'EL +Ve' and 'EL -Ve' indicators.

Above GUI is Test application for GT 5 axis motion control card. Application includes all function of 5 axis motion control card. Detail of application mentioned below.

Default Setting

The value which will be entered here will be applicable for all 5 Axis.



A toolbar for the Default Setting window. It contains a small icon on the left, followed by input fields for 'Acc.Time' (0.1, .1), 'Dec.Time' (0.1, .1), 'High Speed' (50000, 50000), and 'Low Speed' (10000, 10000). On the right are three buttons: 'Set Def.', 'Read Def.', and 'Other Settings'.

Acc. Time: Value for acceleration time in second.

Dec Time: Value for deceleration time in second.

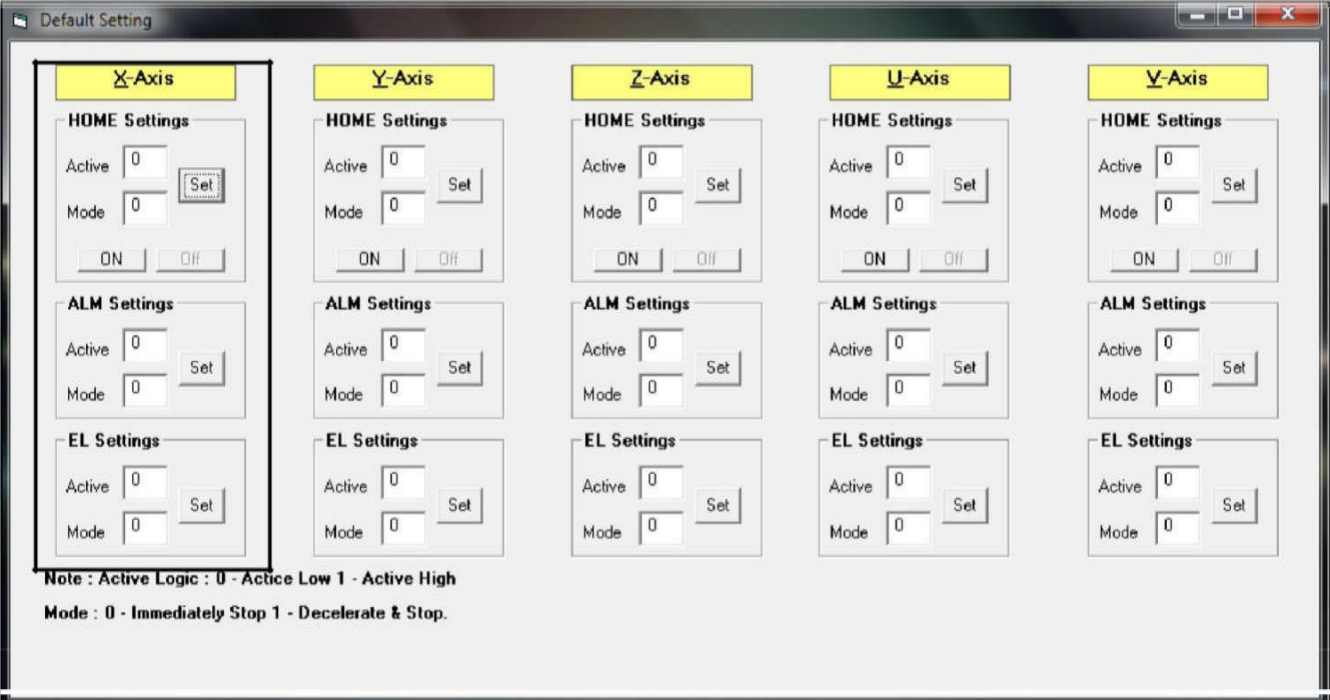
High Speed: Value for maximum velocity in pulse per second (PPS)

Low Speed: Value for starting velocity in pulse per second (PPS)

Set Def: Set entered value

Read Def: Read set value

Other Settings: Window to set active logic and mode of different parameters for a particular axis.



A screenshot of the 'Default Setting' window. It features five columns, each representing an axis: X-Axis, Y-Axis, Z-Axis, U-Axis, and V-Axis. Each column contains three sections: 'HOME Settings', 'ALM Settings', and 'EL Settings'. Each section has 'Active' and 'Mode' dropdown menus, a 'Set' button, and 'ON'/'OFF' toggle buttons. A note at the bottom explains the logic: 'Note : Active Logic : 0 - Active Low 1 - Active High' and 'Mode : 0 - Immediately Stop 1 - Decelerate & Stop.' The X-Axis column is highlighted with a red border.

The screenshot shows a software interface for configuring the X-Axis. It has a yellow header labeled 'X-Axis'. Below it are three sections: 'HOME Settings', 'ALM Settings', and 'EL Settings'. Each section contains 'Active' and 'Mode' dropdown menus, both currently set to '0', and a 'Set' button. The 'HOME Settings' section also includes 'ON' and 'Off' buttons.

- **HOME Setting:** Setting for Origin or home Signal
 - **Active:** Set active logic for ORG.
 - **Mode:** Select reaction mode while detection of ORG signal.
 - **On:** Enable ORG signal. (ORG signal will work as End limit switch)
 - **Off:** Disable ORG signal. (ORG signal will detect only in home function)
 - **Set:** Set above parameter.
- **ALM Setting:** Setting for Alarm signal
 - **Active:** Set active logic for ALM signal.
 - **Mode:** Select reaction mode while detection of ALM signal.
 - **Set:** Set above parameter.
- **EL Setting:** Setting for End limit signal(ELP and ELM)
 - **Active:** Set active logic for ELP and ELM.
 - **Mode:** Select reaction mode while detection of ELP or ELM signal.
 - **Set:** Set Above parameter.

Note:-

- The functions described for X axis as mentioned above will be similarly applicable for other axis.
- Active logic : 0 – Active low (Default)
1 – Active high
- Mode: 0 – Immediately stop (Default)
1 – Deceleration and stop

Individual Axis

X-Axis

H/W Pos. Limit:

S/W Pos. Limit: +Ve -Ve

Speed Limit:

0 Move Count

Vel.

Time

HOME EscHM

JOG

Note: The functions described for X axis as mentioned above will be similarly applicable for other axis.

- **H/W Pos. Limit:** Set hardware position end limit (ELP and ELM)
 - **On:** Will enable the hardware end limit. (ELP and ELM)
 - **Off:** Will disable the hardware end limit. (ELP and ELM)
- **S/W Pos. Limit:** Set end limit by software.
 - **On:** Enables software end limit.
 - **Off:** Disable software end limit.
 - **+Ve:** End limit for positive side. (value should be in pulse)
 - **-Ve:** End limit for negative side. (value should be in pulse)
 - **Set:** To set the values of both limits.
- **Speed Limit:** Set the maximum speed limit
 - **On:** Enable speed limit. (speed value in pulse per second)
 - **Off:** Disable speed limit. (speed value in pulse per second)
 - **Set:** To set the maximum speed limit value
- **Move Count:** Enter value to move relative or absolute position. (in pulse) [+ve count value for +ve direction and -ve count value for -ve direction]
 - **S-A:** S-curve motion profile with absolute position.
 - **S-R:** S-curve motion profile with relative distance
 - **T-R:** Trapezoidal motion profile with relative distance
 - **T-A:** Trapezoidal motion profile with absolute position
- **Read X:** Read current position of X axis. (in pulse)
- **Reset X Count:** Reset/Set X axis position. (in pulse)
- **On Fly Vel. Chg:** Enter the value to change velocity while axis in motion
 - **Vel:** Enter the velocity value in pulse per second
 - **Time:** Enter the acceleration/deceleration time in second
- **Read:** Read current maximum velocity.
- **On Fly Pos. Chg:** Enter the value to change the absolute position while axis in motion.
- **Stop X:** Stop axis after deceleration.
- **Abort:** Stop axis immediately
- **Home:** Home move
 - **[+]:** Search home/ORG signal in positive direction
 - **[-]:** Search home/ORG signal in negative direction
- **EscHM:** Escape from the home position
 - **[+]:** Escape from home position in positive direction.
 - **[-]:** Escape from home position in positive direction.
- **JOG:** Keep axis in motion with maximum velocity. (Velocity Mode)
 - **T [+]:** Move axis with trapezoidal motion profile in positive direction
 - **T [-]:** Move axis with trapezoidal motion profile in negative direction
 - **S [+]:** Move axis with S-curve motion profile in positive direction
 - **S [-]:** Move axis with S-curve motion profile in negative direction

Linear Interpolation

High Speed		Low Speed	
Acc. Time		Dec. Time	
Position/Dist			
Select Mode ☒ Relative ☐ Absolute			
Select Profile ☒ Trapezoid ☐ S-Curve			
Axis1 ☐ X ☐ Y ☐ Z ☐ U ☐ V ☒ None	Axis2 ☐ X ☐ Y ☐ Z ☐ U ☐ V ☒ None	Axis3 ☐ X ☐ Y ☐ Z ☐ U ☐ V ☒ None	
Any 2 Axis	Any 3 Axis	XY	ZU

- **High Speed:** Enter value of maximum velocity.
- **Low Speed:** Enter value of starting velocity.
- **Acc. Time:** enter value for acceleration time. (in sec)
- **Dec. Time:** Enter deceleration time. (in sec)
 - **Position/Dist:** Enter the relative distance or absolute position value for any 2/3 axis.
- **Select Mode:** Select position mode.
- **Select Profile:** Select motion profile.
- **Axis1:** Select first axis for linear interpolation
- **Axis2:** Select second axis for linear interpolation
- **Axis3:** Select third axis for linear interpolation (used only when 3 axis interpolation is needed)
- **Any 2 Axis:** Perform linear interpolation for selected **Axis1** and **Axis2**.
- **Any 3 Axis:** Perform 3 axis linear interpolation for selected **Axis1**, **Axis2** and **Axis3**.
- **XY:** Perform linear interpolation for X and Y axis.
- **ZU:** Perform linear interpolation for Z and U axis

Circular Interpolation

Select Direction		Select Mode	
☒ Clockwise		☒ Relative	
☐ Counter Clockwise		☐ Absolute	
Center X	1000	Center Y	0
End X	1800	End Y	600
Velocity	55000	Arc-Any2 XY ZU	

Select Direction: Select the orientation of circular motion

Select Mode: Select the position mode

Center X: Enter the X co-ordinate value for center. (in pulse)

Center Y: Enter the Y co-ordinate value for center. (in pulse)

End X: Enter the X co-ordinate value for end point. (in pulse)

End Y: Enter the Y co-ordinate value for end point. (in pulse)

Velocity: Enter the velocity for circular motion. (in PPS)

Arc-Any2: Perform circular interpolation for selected **Axis1** and **Axis2**.

XY: Perform circular interpolation for **X** and **Y** axis.

ZU: Perform circular interpolation for **Z** and **U** axis.

General Purpose Digital Output

The control panel for General Purpose Digital Output features a grid of 8 checkboxes for DO0 to DO7. To the right of the grid are two buttons: 'Set DO' and 'Read DO'. Further right is a 'BIT' label followed by a text input field. Below the 'BIT' field are three buttons: 'Set', 'Reset', and 'Bit Read'.

Checkbox: Mark/unmark the checkbox to set/reset the outputs DO0 to DO7.

Set DO: Set marked checkbox

Read DO: Read current status of digital outputs.

BIT: Enter the bit number to set/reset (0-7).

Set: Set the particular bit number or digital output.

Reset: Reset the particular bit number or digital output.

Bit Read: Read the current status of particular bit or digital output.

General Purpose Digital Input

The control panel for General Purpose Digital Input features a grid of 8 checkboxes for DI0 to DI7. To the right of the grid is a 'Read DI' button. Below the grid is a 'Bit Read DI' button, followed by a 'BIT' label and a text input field.

Checkbox: Indicates the status of digital inputs i.e. DI0 to DI7. (mark:'1' & unmark:'0')

Read DI: Reads the status of digital input.

Bit Read DI: Reads the particular digital input.

BIT: Enter the bit number or digital input number to read.

Signal Status

	X	Y	Z	U	V		X	Y	Z	U	V
ALARM	●	●	●	●	●	EL +Ve	●	●	●	●	●
HOME	●	●	●	●	●	EL -Ve	●	●	●	●	●



Signal Low



Signal High

Function List and Description

Function for Initialization

Argument

NumCards – Card index number

Function

1. VB6: GT_PROSYS_U3_Initialize(ByRef NumCards As Long) As Long
VB.Net: GT_PROSYS_U3_Initialize(ByRef NumCards As Int32) As Int32

Description: This function will initialize the card which is connected to the computer.

Example: GT_PROSYS_U3_Initialize (0) – Initialize 0th card.

2. VB6 & VB.Net : GT_PROSYS_U3_Close()

Description: This function closes the current card and releases the resources.

Function for Configuration of IO Pins

Argument

AxisNo	: Axis number designated to set logic or mode
BitNo	: Selecting the particular input or output pin. Value varies from 0 to 7.
d_out	: Integer 8-bit value 0 to 255 which will pass to the output port.
SetVal	: Setting the value of the pin. 0 = LOW. 1 = HIGH.
alm_logic	: Setting of active logic for ALARM signal alm_logic = 0, active LOW. alm_logic = 1, active HIGH.
alm_mode	: Reacting modes when receiving ALARM signal. alm_mode = 0, motor immediately stop (Default) alm_mode = 1, motor decelerates then stop
el_logic	: Setting of active logic for EL signal el_logic = 0, active LOW. el_logic = 1, active HIGH.
el_mode	: Reacting modes when receiving EL signal.

`el_mode = 0`, motor immediately stop (Default)
`el_mode = 1`, motor decelerates then stop
home_logic : Action logic configuration for HOME signal
`home_logic = 0`, active low;
`home_logic = 1`, active high

home_mode : Reacting modes when receiving HOME signal.
`home_mode = 0`, motor immediately stop (Default)
`home_mode = 1`, motor decelerates then stop

Function

3. **VB6** : `GT_PROSYS_U3_get_di()` As Integer
VB.Net: `GT_PROSYS_U3_get_di()` As Int16

Description: This function returns current status of whole digital input port. The function returns the value between 0-255.

4. **VB6** : `GT_PROSYS_U3_get_di_bit(ByVal BitNo As Integer)` As Integer
VB.Net: `GT_PROSYS_U3_get_di_bit(ByVal BitNo As Int16)` As Int16

Description: This function returns the status of the particular input pin(DI) ranges from DI0 to DI7. The function returns the value 0 for low level and 1 for the high level.

Example: `GT_PROSYS_U3_get_di_bit(1)` – Get the status of input pin DI1.

5. **VB6** : `GT_PROSYS_U3_get_do()` As Integer
VB.Net: `GT_PROSYS_U3_get_do()` As Int16

Description: This function returns current status of whole 8-pin digital output port as integer.

Example: `GT_PROSYS_U3_get_do()` – Get the status of all digital output pin (DO0 to DO7) as integer.

6. **VB6** : `GT_PROSYS_U3_get_do_bit(ByVal BitNo As Integer)` As Integer
VB.Net: `GT_PROSYS_U3_get_do_bit(ByVal BitNo As Int16)` As Int16

Description: This function returns the status of the particular output pin ranges from DO0 to DO7. The function returns the value 0 for low level and 1 for the high level.

Example: GT_PROSYS_U3_get_do_bit(0) - Get the status of output pin DO0.

7. VB6 : GT_PROSYS_U3_set_do(ByVal d_out As Integer)
VB.Net: GT_PROSYS_U3_set_do(ByVal d_out As Int16)

Description: This function sets the digital output port DO0 to DO7. D_out value should be in integer between 0-255.

Example: GT_PROSYS_U3_set_do(255) - All the digital output pins to high level.

8. VB6 : GT_PROSYS_U3_set_do_bit(ByVal BitNo As Integer, ByVal SetVal As Integer)
VB.Net: GT_PROSYS_U3_set_do_bit(ByVal BitNo As Int16, ByVal SetVal As Int16)

Description: This function uses to set the particular output pin as high or low.

0 - Set Pin low

1 - Set Pin high

Example: GT_PROSYS_U3_set_do_bit(0,1) – Set output pin DO0 to high.

9. VB6 : GT_PROSYS_U3_get_alm(ByVal AxisNo As Integer) As Integer
VB.Net: GT_PROSYS_U3_get_alm(ByVal AxisNo As Int16) As Int16

Description: This function returns status of the alarm (ALM) input pin of particular axis. Alarm input pin high function will return 1, alarm input pin low function will return 0.

Example: GT_PROSYS_U3_get_alm(0) – Get the status of X-axis alarm signal(xALM)

10. VB6 : GT_PROSYS_U3_set_alm(ByVal AxisNo As Integer, ByVal alm_logic As Integer, ByVal alm_mode As Integer) As Integer

VB.Net: GT_PROSYS_U3_set_alm(ByVal AxisNo As Int16, ByVal alm_logic As Int16, ByVal alm_mode As Int16) As Int16

Description: This function uses to set active logic of alarm signal (ALM) and reacting mode when detect alarm signal.

alm_logic = 0 – Active Low (Default)

alm_logic = 1 – Active High

alm_mode = 0 – Immediately stop (Default)

alm_mode = 1 – Decelerate and stop

Example: GT_PROSYS_U3_set_alm(0, 1, 0) – Set active logic high and reacting mode 0 for X-axis alarm signal.

11. VB6 : GT_PROSYS_U3_get_elp_status (ByVal AxisNo As Integer) As Integer
VB.Net: GT_PROSYS_U3_get_elp_status (ByVal AxisNo As Int16) As Int16

Description: This function returns status of the positive end limit(ELP) input pin of particular axis. ELP input pin high function will return 1, ELP input pin low function will return 0.

Example: GT_PROSYS_U3_get_elp_status (0) – Get the status of X-axis ELP signal(xELP)

12. VB6 : GT_PROSYS_U3_get_eln_status (ByVal AxisNo As Integer) As Integer
VB.Net: GT_PROSYS_U3_get_eln_status (ByVal AxisNo As Int16) As Int16

Description: This function returns status of the negative end limit(ELM) input pin of particular axis. ELM input pin high function will return 1, ELM input pin low function will return 0.

Example: GT_PROSYS_U3_get_eln_status (0) – Get the status of X-axis ELM signal(xELM)

13. VB6 : GT_PROSYS_U3_set_el (ByVal AxisNo As Integer, ByVal el_logic As Integer, ByVal el_mode As Integer) As Integer

VB.Net: GT_PROSYS_U3_set_el(ByVal AxisNo As Int16, ByVal el_logic As Int16, ByVal el_mode As Int16) As Int16

Description: This function uses to set active logic of end limit signal (ELP and ELM) and reacting mode when detect end limit signal.

el_logic	= 0 – Active Low (Default)
el_logic	= 1 – Active High
el_mode	= 0 – Immediately stop (Default)
el_mode	= 1 – Decelerate and stop

Example: GT_PROSYS_U3_set_el(0, 1, 0) – Set active logic high and reacting mode 0 for X-axis end limit signal (xEL+ and xEL-).

14. VB6 : GT_PROSYS_U3_enable_hard_limit(ByVal AxisNo As Integer) As Integer
VB.Net: GT_PROSYS_U3_enable_hard_limit(ByVal AxisNo As Int16) As Int16

Description: This function uses to enable hardware end limit (ELP and ELM) for particular axis.

Example: GT_PROSYS_U3_enable_hard_limit(0) – Enable hardware end limit for X-axis.

15. VB6 : GT_PROSYS_U3_disable_hard_limit(ByVal AxisNo As Integer) As Integer
VB.Net: GT_PROSYS_U3_disable_hard_limit(ByVal AxisNo As Int16) As Int16

Description: This function uses to disable hardware end limit (ELP and ELM) for particular axis.

Example: GT_PROSYS_U3_disable_hard_limit(0) – Disable hardware end limit for X-axis.

16. VB6 : GT_PROSYS_U3_get_home_status (ByVal AxisNo As Integer) As Integer
VB.Net: GT_PROSYS_U3_get_home_status (ByVal AxisNo As Int16) As Int16

Description: This function returns status of the Home or Origin (ORG) input pin of particular axis. ORG input pin high function will return 1, ORG input pin low function will return 0.

Example: GT_PROSYS_U3_get_home_status (0) – Get the status of X-axis ORG signal(xORG)

17. VB6 : GT_PROSYS_U3_set_home (ByVal AxisNo As Integer, ByVal home_logic As Integer, ByVal home_mode As Integer) As Integer

VB.Net: GT_PROSYS_U3_set_home(ByVal AxisNo As Int16, ByVal home_logic As Int16, ByVal home_mode As Int16) As Int16

Description: This function uses to set active logic of origin or home signal (ORG) and reacting mode when detect origin signal.

home_logic = 0 – Active Low (Default)

home_logic = 1 – Active High

home_mode = 0 – Immediately stop (Default)

home_mode = 1 – Decelerate and stop

Example: GT_PROSYS_U3_set_home(0, 1, 0) – Set active logic high and reacting mode 0 for X-axis origin signal (xORG).

Function For Velocity Mode Motion

Argument

AxisNo	: Axis number designated to move or stop.
StrVel	: Starting velocity in unit of pulse per second
MaxVel	: Maximum velocity in unit of pulse per second
Tacc	: Specified acceleration time in unit of second
NewVel	: New velocity in unit of pulse per second
Tdec	: Specified deceleration time in unit of second
TimeSecond	: Acceleration/deceleration time for on the fly speed change
Dir	: Set the direction of the motor 0 = Clock wise direction 1 = Counter clock wise direction

Function

15. VB6 : GT_PROSYS_U3_tv_move(ByVal AxisNo As Integer, ByVal StrVel As Double, ByVal MaxVel As Double, ByVal Tacc As Double, ByVal dir As Integer) As Integer

VB.Net: GT_PROSYS_U3_tv_move(ByVal AxisNo As Int16, ByVal StrVel As Double, ByVal MaxVel As Double, ByVal Tacc As Double, ByVal dir As Int16) As Int16

Description: This function uses for accelerate any axis from starting velocity to maximum velocity with the trapezoidal profile in specified direction. By calling this function axis will continuously move at maximum speed. It will stop only when we call the stop or abort function.

Example: GT_PROSYS_U3_tv_move(0, 1000, 5000, 0.1, 0) – By calling this function X-axis accelerate from starting velocity 1000 pulse per second(PPS) to maximum velocity 5000 PPS in 0.1 second with trapezoidal motion profile and move with maximum velocity in 0 (Clock wise) direction until stop or abort command issue.

16. VB6 : GT_PROSYS_U3_v_change(ByVal AxisNo As Integer, ByVal NewVel As Double, ByVal TimeSecond As Double) As Integer

VB.Net: GT_PROSYS_U3_v_change(ByVal AxisNo As Int16, ByVal NewVel As Double, ByVal TimeSecond As Double) As Int16

Description: This function changes velocity on the fly with trapezoidal or s-curve motion profile. In this function the existing velocity and the new velocity will be compared and the time will act as the acceleration and the deceleration time accordingly.

Example: GT_PROSYS_U3_v_change(0, 50000, 0.1) – By calling this function X-axis velocity reaches to 50000 PPS in 0.1 second from current velocity.

17. VB6 : GT_PROSYS_U3_sd_stop(ByVal AxisNo As Integer, ByVal Tdec As Double) As Integer

VB.Net: GT_PROSYS_U3_sd_stop(ByVal AxisNo As Int16, ByVal Tdec As Double) As Int16

Description: This function is used to decelerate an axis to starting velocity with trapezoidal profile or S-curve profile and stop. Velocity profile is decided by original motion profile.

Example: GT_PROSYS_U3_sd_stop(0, 0.1) – By calling this function X-axis start decelerate to starting velocity with trapezoidal or s-curve motion profile in 0.1 second and stop.

18. VB6 : GT_PROSYS_U3_abort(ByVal AxisNo As Integer) As Integer

VB.Net: GT_PROSYS_U3_abort(ByVal AxisNo As Int16) As Int16

Description: This function is used to immediately stop an axis.

Example: GT_PROSYS_U3_abort(0) – By calling this function X-axis stop immediately.

19. VB6 : GT_PROSYS_U3_fix_speed_range(ByVal AxisNo As Integer, ByVal MaxVel As Double) As Integer

VB.Net: GT_PROSYS_U3_fix_speed_range(ByVal AxisNo As Int16, ByVal MaxVel As Double) As Int16

Description: This function is used to set maximum speed range. By setting speed limit on particular axis, axis high speed will not go beyond this speed (Speed in Pulse per second).

Example: GT_PROSYS_U3_fix_speed_range(0, 10000) – By calling this function X-axis maximum speed set to 10000 PPS.

20. VB6 : GT_PROSYS_U3_enable_speed_limit(ByVal AxisNo As Integer) As Integer

VB.Net: GT_PROSYS_U3_enable_speed_limit(ByVal AxisNo As Int16) As Int16

Description: This function is used to enable speed limit on selected Axis.

Example: GT_PROSYS_U3_enable_speed_limit(0) – Enable speed limit on X-axis.

21. VB6 : GT_PROSYS_U3_disable_speed_limit(ByVal AxisNo As Integer) As Integer

VB.Net: GT_PROSYS_U3_disable_speed_limit(ByVal AxisNo As Int16) As Int16

Description: This function is used to disable speed limit on selected Axis.

Example: GT_PROSYS_U3_disable_speed_limit(0) – Disable speed limit on X-axis.

22. VB6 : GT_PROSYS_U3_get_axis_vel(ByVal AxisNo As Integer) As Double

VB.Net: GT_PROSYS_U3_get_axis_vel(ByVal AxisNo As Int16) As Double

Description: This function is used to get current velocity of particular axis.

Example: GT_PROSYS_U3_get_axis_vel(0) – Get current speed of X-axis.

Function for Motion

VB.Net : GT_PROSYS_U3_axis_move (ByVal Rel_Abs as Int16 , ByVal NoAxis as Int16, ByRef AxisArray as Int16, ByRef DistArray as double, ByVal StrVel as double, ByVal MaxVel as double, ByVal Tacc as double);

VB6 : GT_PROSYS_U3_axis_move (ByVal Rel_Abs as Integer, ByVal NoAxis as Integer, ByRef AxisArray as Integer, ByRef DistArray as double, ByVal StrVel as double, ByVal MaxVel as double, ByVal Tacc as double);

Argument:

Rel_Abs : Absolute or Relative Movement

for Relative = 0

Absolute = 1

NoAxis : No. of Axis to Move

ex. : 1 :single axes to move
2 :interpolation of 2 axis
3 :interpolation of 3 axis
4 :Arc/Circular Interpolation

AxisArray[] : Which axis to move

ex. Move X axis only , AxisArray[0] = 0

ex. Interpolation of X and Z axis , AxisArray[0] = 0

AxisArray[1] = 2

ex. Interpolation of Y, Z and U axis ,

AxisArray[0] = 1

AxisArray[1] = 2

AxisArray[2] = 3

ex. arc/circular interpolation of X and Y axis,

AxisArray[0] = 0

AxisArray[1] = 1

DistArray[] : No. of Pulses to Move

ex. Move X axis to 10000 pulses,

DistArray[0]=10000

ex. Interpolation of Z and V axis with Pulses 1000 and 33000,

DistArray[0] = 1000

DistArray[1] = 33000

ex. Interpolation of X,Z and V axis with Pulses 2000, 15000 and 25000,

DistArray[0] = 2000

DistArray[1] = 15000

DistArray[2] = 25000

ex. arc/circular interpolation

DistArray[0] = cx

DistArray[1] = cy

DistArray[2] = ex

DistArray[3] = ey

DistArray[4] = direction

StrVel : Starting Velocity

MaxVel : Maximum Velocity
Tacc : Acceleration Time in Second

Function

24. Move X axis to absolute position 10000, with starting velocity 1000 and Maximum Velocity 5000, and Acceleration time 0.1 Second

```
rel_abs = 1;  
Total_axis_to_move = 1;  
double str_vel = 1000;  
double max_vel = 5000;  
double acc = 0.1;
```

```
axis_array[0] = 0;  
position_array[0] = 10000;
```

```
fixed (double* position = position_array) fixed (Int16* axis = axis_array)  
GT_PROSYS_U3_axis_move(rel_abs, Total_axis_to_move, axis, position, str_vel, max_vel, acc);
```

25. Move Z axis to relative position 15000, with starting velocity 5000 and Maximum Velocity 10000, and Acceleration time 0.05 Second

```
rel_abs = 0;  
Total_axis_to_move = 1;  
double str_vel = 5000;  
double max_vel = 10000;  
double acc = 0.05;
```

```
axis_array[0] = 2;  
position_array[0] = 15000;
```

```
fixed (double* position = position_array) fixed (Int16* axis = axis_array)  
GT_PROSYS_U3_axis_move(rel_abs, Total_axis_to_move, axis, position, str_vel, max_vel, acc);
```

26. Move/Interpolate Y and U axis to relative position 2000 and 5000, with starting velocity 1000 and Maximum Velocity 5000, and Acceleration time 0.1 Second

```
rel_abs = 0;  
Total_axis_to_move = 2;  
double str_vel = 1000;  
double max_vel = 5000;  
double acc = 0.1;
```

```
axis_array[0] = 1;  
axis_array[1] = 3;
```

```
position_array[0] = 2000;
```

```
position_array[1] = 5000;
```

```
fixed (double* position = position_array) fixed (Int16* axis = axis_array)
```

```
GT_PROSYS_U3_axis_move(rel_abs, Total_axis_to_move, axis, position, str_vel, max_vel, acc);
```

27. Move X and V axis to absolute position 10000 and 50000, with starting velocity 15000 and Maximum Velocity 25000, and Acceleration time 0.2 Second

```
rel_abs = 1;
```

```
Total_axis_to_move = 2;
```

```
double str_vel = 15000;
```

```
double max_vel = 25000;
```

```
double acc = 0.2;
```

```
axis_array[0] = 0;
```

```
axis_array[1] = 4;
```

```
position_array[0] = 10000;
```

```
position_array[1] = 50000;
```

```
fixed (double* position = position_array) fixed (Int16* axis = axis_array)
```

```
GT_PROSYS_U3_axis_move(rel_abs, Total_axis_to_move, axis, position, str_vel, max_vel, acc);
```

28. Draw ARC with Cx=0,Cy=1000,Ex=0,Ey=2000 absolute position ,Maximum Velocity 5000 and Accelration Time 0.1 seconds

```
rel_abs = 1;
```

```
Total_axis_to_move = 4;
```

```
double max_vel = 5000;
```

```
double acc = 0.1;
```

```
axis_array[0] = 0;
```

```
axis_array[1] = 1;
```

```
position_array[0] = 0;
```

```
position_array[1] = 1000;
```

```
position_array[2] = 0;
```

```
position_array[3] = 2000;
```

```
position_array[4] = 1;
```

```
fixed (double* position = position_array) fixed (Int16* axis = axis_array)
```

```
GT_PROSYS_U3_axis_move(rel_abs, Total_axis_to_move, axis, position, str_vel, max_vel, acc);
```

29. VB6 : GT_PROSYS_U3_p_change(ByVal AxisNo As Integer, ByVal NewPos As Double)
As Integer

VB.Net: GT_PROSYS_U3_p_change(ByVal AxisNo As Int16, ByVal NewPos As Double)
As Int16

Description: This function is used to change target position on the fly. After changing position, the axis will move to the new target position and totally forget the original position. If the new position is in passed path, it will cause the axis to decelerate to stop, than reverse the direction and move to new position. The acceleration and deceleration rate, the starting velocity and maximum velocity will keep the same as original setting.

Example: GT_PROSYS_U3_p_change(0, 50000)

30. VB6 : GT_PROSYS_U3_get_command(ByVal AxisNo As Integer, ByRef cmd As Long) As Integer

VB.Net: GT_PROSYS_U3_get_command(ByVal AxisNo As Int16, ByRef cmd As Int32) As Int16

Description: This function is used to get the current position of the particular axis.

Example: GT_PROSYS_U3_get_command(0,cmd), cmd = <Value>

31. VB6 : GT_PROSYS_U3_set_command(ByVal AxisNo As Integer, ByVal cmd As Long) As Integer

VB.Net: GT_PROSYS_U3_set_command(ByVal AxisNo As Int16, ByVal cmd As Int32) As Int16

Description: This function is used to set the position of the particular axis.

Example: GT_PROSYS_U3_set_command(0, 1000)

Function For Home Return Mode

Argument

AxisNo	:	Axis number designated to set logic or mode and home move
home_logic	:	Action logic configuration for HOME signal home_logic= 0, active low; home_logic = 1, active high
home_mode	:	Reacting modes when receiving HOME signal. home_mode = 0, Immediate Stop home_mode = 1, Slow down then Stop
StrVel	:	Starting velocity of a velocity profile in unit of pulse per second
MaxVel	:	Maximum velocity of a velocity profile in unit of pulse per second
Tacc	:	Specified acceleration time in unit of second
Tdec	:	Specified deceleration time in unit of second
Dir	:	Set the direction of the motor 0 = Clock wise direction 1 = Counter clock wise direction

Function

32. VB6 : GT_PROSYS_U3_set_home (ByVal AxisNo As Integer, ByVal home_logic As Integer, ByVal home_mode As Integer) As Integer

VB.Net: GT_PROSYS_U3_set_home(ByVal AxisNo As Int16, ByVal home_logic As Int16, ByVal home_mode As Int16) As Int16

Description: This function uses to set active logic of origin or home signal (ORG) and reacting mode when detect origin signal.

home_logic = 0 – Active Low (Default)
home_logic = 1 – Active High
home_mode = 0 – Immediately stop (Default)
home_mode = 1 – Decelerate and stop

Example: GT_PROSYS_U3_set_home(0, 1, 0) – Set active logic high and reacting mode 0 for X-axis origin signal (xORG).

33. VB6 : GT_PROSYS_U3_home_axis (ByVal AxisNo As Integer, ByVal StrVel As Double, ByVal MaxVel As Double, ByVal Tacc As Double, ByVal dir As Integer) As Integer

VB.Net: GT_PROSYS_U3_home_axis(ByVal AxisNo As Int16, ByVal StrVel As Double, ByVal MaxVel As Double, ByVal Tacc As Double, ByVal dir As Int16) As Int16

Description: This function will perform home move for selected axis. Axis will only stop at detection of ORG or home signal. User can stop axis by calling stop or abort function.

Example: GT_PROSYS_U3_home_axis(0, 1000, 50000, 0.1, 0)

34. VB6 : GT_PROSYS_U3_escape_home(ByVal AxisNo As Integer, ByVal StrVel As Double, ByVal MaxVel As Double, ByVal Tacc As Double, ByVal dir As Integer) As Integer

VB.Net: GT_PROSYS_U3_escape_home(ByVal AxisNo As Int16, ByVal StrVel As Double, ByVal MaxVel As Double, ByVal Tacc As Double, ByVal dir As Int16) As Int16

Description: Function use to escape from home position. This function will work only when home or ORG signal is detect.

Example: GT_PROSYS_U3_escape_home(0, 1000, 5000, 0.1, 0)

Function for Motion Status

Argument

Axis No. : Axis No. Designated to Monitor Motion

Function

35. VB6 : GT_PROSYS_U3_motion_done(ByVal AxisNo As Integer) As Integer
VB.Net: GT_PROSYS_U3_motion_done(ByVal AxisNo As Int16) As Int16

Description: This function is used to monitor the particular axis which is currently in a motion or not. If axis is in motion, it will only return command to the program when motion will complete.

Example: GT_PROSYS_U3_motion_done(0) – Check X axis motion complete or not.

- VB6: GT_PROSYS_U3_all_motion_status() As Integer
VB.Net: GT_PROSYS_U3_all_motion_status() As INT16

Description: This function will return all axis status as word.

Function for motion interface input/output

Argument

AxisNo	: Axis number designated to set logic or mode
alm_logic	: Setting of active logic for ALARM signal alm_logic = 0, active LOW (Default) alm_logic = 1, active HIGH
alm_mode	: Reacting modes when receiving ALARM signal alm_mode = 0, motor immediately stop (Default) alm_mode = 1, motor decelerates then stop
el_logic	: Setting of active logic for EL signal el_logic = 0, active LOW (Default) el_logic = 1, active HIGH
el_mode	: Reacting modes when receiving EL signal. el_mode = 0, motor immediately stop (Default) el_mode = 1, motor decelerates then stop

Function

36. VB6 : GT_PROSYS_U3_set_alm(ByVal AxisNo As Integer, ByVal alm_logic As Integer, ByVal alm_mode As Integer) As Integer

VB.Net: GT_PROSYS_U3_set_alm(ByVal AxisNo As Int16, ByVal alm_logic As Int16, ByVal alm_mode As Int16) As Int16

Description: This function uses to set active logic of alarm signal (ALM) and reacting mode when detect alarm signal.

alm_logic = 0 – Active Low (Default)
alm_logic = 1 – Active High
alm_mode = 0 – Immediately Stop (Default)
alm_mode = 1 – Decelerate and Stop

Example: GT_PROSYS_U3_set_alm(0, 1, 0) – Set active logic high and reacting mode 0 for X-axis alarm signal.

37. VB6 : GT_PROSYS_U3_set_el (ByVal AxisNo As Integer, ByVal el_logic As Integer, ByVal el_mode As Integer) As Integer

VB.Net: GT_PROSYS_U3_set_el(ByVal AxisNo As Int16, ByVal el_logic As Int16, ByVal el_mode As Int16) As Int16

Description: This function uses to set active logic of end limit signal (ELP and ELM) and reacting mode when detect end limit signal.

el_logic = 0 – Active Low (Default)
el_logic = 1 – Active High
el_mode = 0 – Immediately stop (Default)
el_mode = 1 – Decelerate and stop

Example: GT_PROSYS_U3_set_el(0, 1, 0) – Set active logic high and reacting mode 0 for X-axis end limit signal (xEL+ and xEL-).

38. VB6 : GT_PROSYS_U3_enable_hard_limit(ByVal AxisNo As Integer) As Integer
VB.Net: GT_PROSYS_U3_enable_hard_limit(ByVal AxisNo As Int16) As Int16

Description: This function uses to enable hardware end limit (ELP and ELM) for particular axis.

Example: GT_PROSYS_U3_enable_hard_limit(0) – Enable hardware end limit for X-axis.

39. VB6 : GT_PROSYS_U3_disable_hard_limit(ByVal AxisNo As Integer) As Integer
VB.Net: GT_PROSYS_U3_disable_hard_limit(ByVal AxisNo As Int16) As Int16

Description: This function uses to disable hardware end limit (ELP and ELM) for particular axis.

Example: GT_PROSYS_U3_disable_hard_limit(0) – Disable hardware end limit for X-axis.

Function for Position Control

Argument

AxisNo : Axis number designated to set or get position
Cmd : Variable which store position value

Function

40. VB6 : GT_PROSYS_U3_get_command(ByVal AxisNo As Integer, ByRef cmd As Integer Long) As

VB.Net: GT_PROSYS_U3_get_command(ByVal AxisNo As Int16, ByRef cmd As Int32) As Int16

Description: This function is used to get the current position of the particular axis.

Example: GT_PROSYS_U3_get_command(0, 35600)

41. VB6 : GT_PROSYS_U3_set_command(ByVal AxisNo As Integer, ByVal cmd As Long) As Integer

VB.Net: GT_PROSYS_U3_set_command(ByVal AxisNo As Int16, ByVal cmd As Int32) As Int16

Description: This function is used to set the position of the particular axis.

Example: GT_PROSYS_U3_set_command(0, 1000)

Function for Position Compare

Argument

AxisNo : Axis number designated to set position, enable or disable limit
PLimit : Software limit value in positive direction
NLimit : Software limit value in negative direction

Function

42. VB6 : GT_PROSYS_U3_set_soft_limit(ByVal AxisNo As Integer, ByVal PLimit As Long, ByVal NLimit As Long) As Integer

VB.Net: GT_PROSYS_U3_set_soft_limit(ByVal AxisNo As Int16, ByVal PLimit As Int32, ByVal NLimit As Int32) As Int16

Description: This function use to set position limit by software. Here, PLimit – Positive position limit and NLimit – Negative position limit.

Example: GT_PROSYS_U3_set_soft_limit(0, 150000, -150000)

43. VB6 : GT_PROSYS_U3_enable_soft_limit(ByVal AxisNo As Integer) As Integer
VB.Net: GT_PROSYS_U3_enable_soft_limit(ByVal AxisNo As Int16) As Int16

Description: Enable Soft limit function on selected axis.

Example: GT_PROSYS_U3_enable_soft_limit(0)

43. VB6 : GT_PROSYS_U3_disable_soft_limit(ByVal AxisNo As Integer) As Integer
VB.Net: GT_PROSYS_U3_disable_soft_limit(ByVal AxisNo As Int16) As Int16

Description: Disable Soft limit function on selected axis.

Example: GT_PROSYS_U3_disable_soft_limit(0)

Function for Continuous motion

Function

44. GT_PROSYS_U3_set_continuous_mode_on()

Description: This function will set card in continuous mode.

45. GT_PROSYS_U3_set_continuous_mode_off()

Description: This function will set card in normal mode.

46. GT_PROSYS_U3_buffer_maxsize() As Integer

Description: This function will return maximum buffer size of card as integer.

47. GT_PROSYS_U3_get_buffer_size() As integer

Description: This function will return current buffer size of card as integer.

48. GT_PROSYS_U3_buffer_ready() As integer

Description: This function will return 0 (zero) if card buffer is ready and one if card buffer is not ready.

49. GT_PROSYS_U3_send_buffer()

Description: This function will send the filled buffer for execution.

50. GT_PROSYS_U3_buf_abort()

Description: This function will abort buffer execution and set card to normal mode.

51. GT_PROSYS_U3_get_buf_cur_id(ByRef CMDId As Integer)

Description: This function will return current command number which is in execution.

Function for Galvo

Function

52. GT_PROSYS_U3_galvo_run_status()

Description: This function will return galvo status

1: Galvo is running

0: Galvo is stopped

53. GT_PROSYS_U3_get_XY_Data(ByRef Data1 Int32, ByRef Data2 Int32)

Description: This function will return galvo position

Data1: x-Coordinate

Data2: y-Coordinate

54. GT GT_PROSYS_U3_XY_Data(ByVal val1 As Int16 ByVal val1 As UInt32)

Description: This function is used to add axis and its position

val1: 0-x axis, 1-y axis

val2: Axis Coordinate

55. GT_PROSYS_U3_Run_Galvo(ByVal val1 As Int16)

Description: This function is used to run galvo as per added axis and its position

1: To run galvo

56. GT_PROSYS_U3_galvo_move_XY(ByVal x2 As UInt16 ByVal Y2 As UInt16, ByVal
gspeed As Int32, ByVal XOFFSET As Double, ByVal YOFFSET As Double, ByVal THETA
As Int16)

Description: This function is used to run galvo as per added axis and its position.

This function is also supported by buffer mode

x2: X-Coordinate

y2: Y-Coordinate,

gspeed: galvo speed maximum up to 1000000 points/sec

XOFFSET: OFFSET Added to X-Coordinate,

YOFFSET: OFFSET Added to Y-Coordinate,

THETA: PASS 0(future development)

57. GT_PROSYS_U3_galvo_stop()

Description: This function is used Stop Galvo

Function for Memory

Function

58. GT_PROSYS_U3_get_mem (ByRef b1 As byte, ByRef b2 As byte, ByRef b3 As byte,
ByRef b4 As byte, ByRef b5 As byte, ByRef b6 As byte, ByRef b7 As byte, ByRef b8 As
byte, ByRef b9 As byte, ByRef b10 As byte);

Description: This function is used to set user defined 10 byte memory

59. GT_PROSYS_U3_set_mem (ByVal b1 As byte, ByVal b2 As byte, ByVal b3 As byte,
ByVal b4 As byte, ByVal b5 As byte, ByVal b6 As byte, ByVal b7 As byte, ByVal b8 As
byte, ByVal b9 As byte, ByVal b10 As byte);

Description: This function is used to get user defined 10 byte memory



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